

Technology Stack

Date	02 July 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, JavaScript	Provides a simple, responsive, and user-friendly interface for entering loan applicant details, validating user inputs, and displaying loan eligibility prediction results effectively.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask handles HTTP requests, processes user inputs, loads the trained machine learning model, performs prediction, and returns loan eligibility results through the web application.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset (.csv), Pickle Model Files (.pkl)	The loan applicant dataset is stored in CSV format for model training and testing. The trained machine learning model and scaler are stored as Pickle (.pkl) files for efficient real-time prediction.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render (Python Web Service)	The Smart Lender application is deployed on Render, enabling users to access the Flask web application online and perform real-time loan eligibility prediction through a web browser.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git, GitHub	Git is used for version control, while GitHub is used to manage the project source code, maintain version history, collaborate with team members, and host the project repository.

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Joblib, Jupyter Notebook	Pandas and NumPy are used for data preprocessing and analysis. Matplotlib and Seaborn are used for data visualization. Scikit-learn is used for preprocessing and model evaluation. XGBoost is used as the final machine learning algorithm for loan prediction. Joblib is used to save and load the trained model, while Jupyter Notebook is used for model development and experimentation.
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